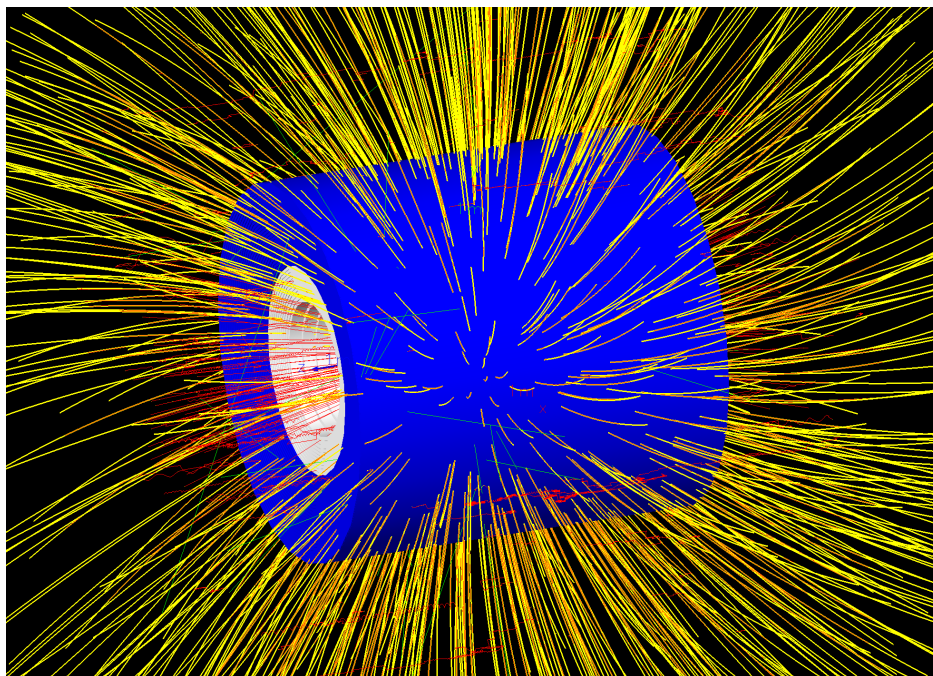


DriftChamberSim Technical Design Document

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MOTIVATION

The Belle II Experiment is particle physics experiment studying the properties of B mesons that operates a detector surrounding the SuperKEKB e^+e^- collider in Tsukuba, Japan with a peak luminosity of $5.319 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$. The current configuration of the accelerator will be unable to meet Belle II's next luminosity upgrade goal of $6.0 \times 10^{35} \text{ cm}^{-2} \text{ s}^{-1}$, and thus a redesign of various parts is necessary. [1, p.2] One of these is the detector's Central Drift Chamber (CDC), the component used to track the momentum of particles resulting from collisions. Proposals for an upgrade to the CDC include changes to the internal gas mixture, a different configuration of axial and stereo wire superlayers, and the addition of cluster counting from a trace of the sense wire signal. To test the potential effects of these changes, it was necessary to create a computer simulation modelling the current CDC with the ability to modify its structure, composition, and behaviour according to the proposed upgrades. The simulation tool *DriftChamberSim* was developed as part of this initiative.

OVERVIEW

Within the drift chamber are alternating superlayers of axial and stereo wires, each composed of multiple wire sublayers. Assuming the chamber is aligned with the z-axis, wires within the axial layers maintain fixed x and y positions while extending along the z-axis while wires within the stereo layers are skewed relative to the layer's radius. Each wire sublayer is divided in equally sized drift cells, made up a single charged sense wire surrounded by eight ground wires.

Following a collision, secondary particles scatter through a drift chamber. An internal magnetic field causes these particles to travel in helical trajectories that cannot be observed directly. The particle ionizes a stable gas mixture along its path, liberating electrons from the surrounding gas atoms. Often these kicked electrons have enough kinetic energy to knock more electrons out of neighbouring atoms, creating a localized burst of secondary ionizations at that point on the particle's track, called a "cluster". The liberated electrons drift toward the nearest sense wire, triggering an electrical pulse upon contact. By combining r - ϕ data from axial wires with z data from stereo wires, the hit's drift cell can be estimated and the particle's trajectory and momentum can be reconstructed to identify the particle type.

PROGRAMMING TOOL

As Cuna notes in *Simulation of particle identification with the cluster counting technique*, "Garfield++ can describe in detail the properties and the performance of a single cell or drift chambers, but it is not suitable to simulate a large scale detector and to study colliders events" [2, p.6]. Similarly, Geant4 is sufficiently suitable for simulating and measuring complex behaviour of particles through a medium, but falters when it comes to smaller-scale behaviour, such as the performance of a single cell, without significant parametrization. Neither program is sufficient for simulating every detail necessary for a robust drift chamber simulation. As such, *DriftChamberSim* uses Geant4 for its main simulation of particle trajectories with supplementary constants derived from the Garfield++ program *DriftTimes*. *DriftTimes* simulates a single, isolated drift cell within a volume of the specified gas mixture. It divides this cell into a $1 \text{ mm} \times 1 \text{ mm}$ grid and runs until a particle cluster is generated within each region. From this particle cluster, it derives the average time it takes for the particles in the cluster to drift to the sense wire, as well as the rate at which the particles can move through the gas medium. Finally, it exports these resulting drift times and diffusion constants to CSV files, creating a lookup table for reference by the main simulation.

GEOMETRY

The geometry of *DriftChamberSim* is centred at the origin, from which particles originate with a direction randomly generated from a uniform distribution. To model the sense wire hit detector, the simulation structures the internal volume as a series of sensitive gas layers, each corresponding to a specific wire sublayer. Rather than modelling individual wires, which would drastically increase computational overhead, the simulation organizes the particle hits by wire layer ID through these gas layers. To account for the material effects of the wires, the simulated gas mixture in each sublayer includes a calculated percentage of the wire material based on the number of wires in that layer.

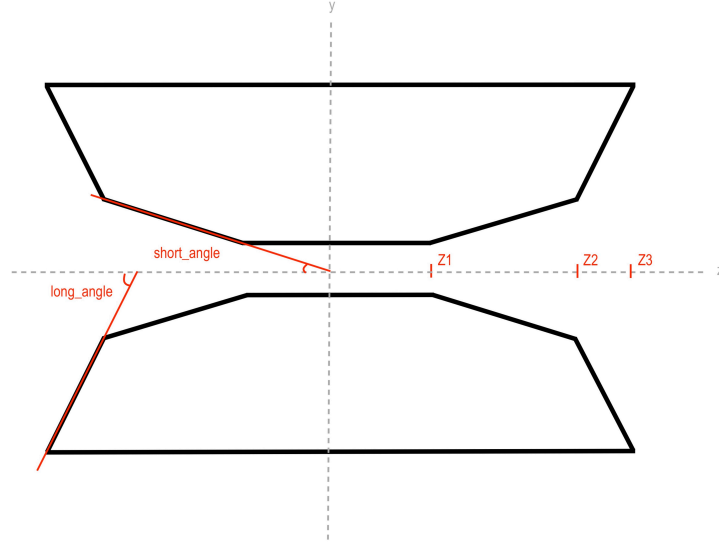


FIG. 1. Diagram of the main structure of the *DriftChamberSim* geometry.

Both ends of the CDC are covered by two sections of tapered endplates made up of overlapping rings [3, p.205]. *DriftChamberSim* defines the detector geometry using three key axial coordinates: Z_1 (the start of the first endplate section), Z_2 (the start of the second endplate section), and Z_3 (the end of the detector). In Geant4, the detector geometry is constructed by starting with a solid annular cylindrical volume and subtracting two conical volumes that correspond to the two endplate sections. As shown in Fig 1, these are modelled as a "short cutout" spanning from Z_1 to Z_2 and a "long cutout" spanning from Z_2 to Z_3 . While the CDC features both endplate regions, *DriftChamberSim* only implements the endplate within the long cutout region.

GAS MIXTURE

The gas mixture inside the drift chamber determines the drift velocity of the ionized electrons and the signal stability needed to reconstruct particle trajectories. The gas volume for the current Belle II CDC consists of an equal-volume mixture of Helium and Ethane ($\text{He:C}_2\text{H}_6$, 50:50 %). *DriftChamberSim* includes nine gas files generated using the Garfield++ Magboltz generator tool that can be used by the Drift-Time generator to simulate the behaviour of the particle through a volume of that mixture. The gases were selected from a list of gas mixtures in similar particle detectors and proposed changes to the current Belle II gas mixture.

The MEG-II Experiment uses a modified He:Isobutane mixture ($\text{He}:\text{C}_4\text{H}_{10}$, 90 : 10 %) with an additional 0.5% O_2 and 1.5% $\text{C}_2\text{H}_6\text{O}$ in their detector. Their research indicates that this addition of small percentages of Isopropyl Alcohol or water and Oxygen is effective at stabilizing the chamber's performance [4]. One CDC proposal suggests this practice be adopted by Belle II.

While Belle II uses a Helium-based gas mixture, Argon-based mixtures are common in other experiments. For one, the ATLAS Experiment uses a Argon based gas mixture ($\text{Ar}:\text{CO}_2$, 93 : 7 %). A 90 : 10 % mixture of Argon and Methane (CH_4), known as P10 gas, is commonly used in Time Projection Chambers for accurate 3D particle tracking. CLEO, a now decommissioned particle detector at the Cornell Electron Storage Ring used for conducting similar studies on B mesons to Belle II, initially utilized a Argon:Ethane mixture ($\text{Ar}:\text{C}_2\text{H}_6$, 60 : 40 %), but later switched to Helium:Propane($\text{He}:\text{C}_3\text{H}_8$, 60 : 40 %) to increase their radiation length [5, p.95].

DELTA RAYS

As a charged particle traverses the drift chamber, it will knock low-energy electrons from the surrounding gas atoms. Occasionally, however, the energy of the kicked electron will be sufficient to travel a significant distance from the primary track. These high-energy electrons, known as "delta rays," induce secondary ionization along their paths, often at an elevated rate due to concentrated energy deposition (see Fig 2.).

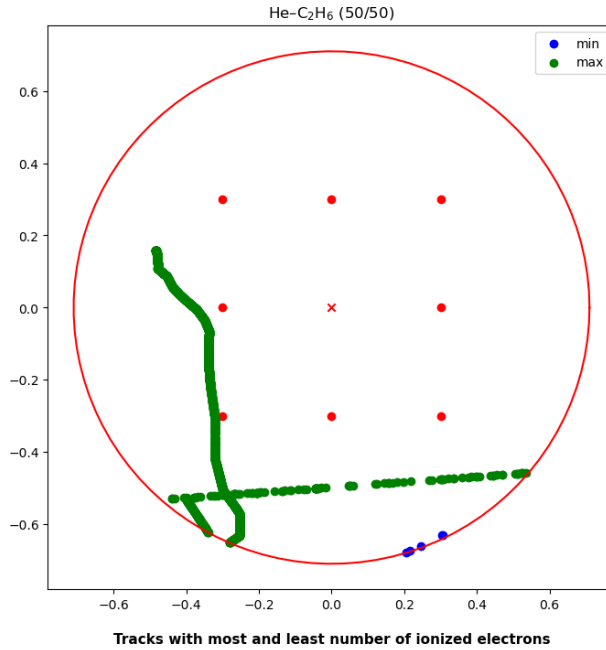


FIG. 2. Simulated electron initial positions for the shortest and longest track lengths from a sample of $n = 1000$ randomly generated tracks in Garfield++.

The phenomenon of delta rays is replicable in Geant4 and Garfield++ and, given that differences between the primary and secondary ionization is significant, it is important to separate the two data sets. In *DriftChamberSim*, hits within layers of the drift chamber were classified as originating from either ions or non-ions based on their process name. Consequently, any ion that deposited enough energy to register a hit was identified as a delta ray.

RECONSTRUCTION

Simulation tools like Garfield++ and Geant4 are useful because of the depths of their ability and accuracy to measure and track the behaviour of particles in the environment being modelled. *DriftChamberSim* tracks the exact trajectory and momentum of the particles through its volume since those particles are generated as a part of the simulation. SuperKEKB can only track the momentum of a particle through the CDC from hits along the sense wires generated by the resulting ionized electrons. To replicate this behaviour, the reconstruction tool *KalmanFit* estimates the uncertainty by using the position on the nearest sense wire closest to each hit.

The wire point of closest approach (**P**) to the actual hit position (**T**) is described by

$$\mathbf{P} = \mathbf{A} + \left(\frac{\mathbf{T} - \mathbf{A}}{\|\mathbf{V}\|^2} \right) \cdot \mathbf{V} \quad (1)$$

where **V** is the direction vector of the wire and **A** is a point on it.

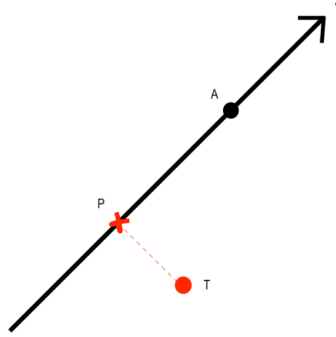


FIG. 3. Diagram of sense wire hit estimation.

Kalman fitting

Given that the CDC only records individual sense wire hits, it is necessary to later reconstruct the particle's trajectory from these hits. Belle II uses a method called Kalman fitting, a process of track estimation that uses recursive noise-filtering to find an average slope, sourced from the GenFit tool [6, p.3]. The *KalmanFit* reconstruction tool uses this same technique with the estimated sense wire hits. Comparing the trajectories of the particles from the simulation and those reconstructed by *KalmanFit* (see Fig. 8. in APPENDIX I), the fitting is generally a sufficient approximation in the x-y plane, with greater deviation in z. Further adjustments to the seeding or a change in fitting method are likely needed to improve the accuracy of this fitting.

Cluster Counting

A signal plot visualizes the electric current induced on readout electrodes over time by ionized electrons drifting toward the connected sense wires. Peaks on the plot correspond to electron clusters originating along the particle track ($\frac{dN}{dx}$), while the area under the curve represents the total energy deposited ($\frac{dE}{dx}$).

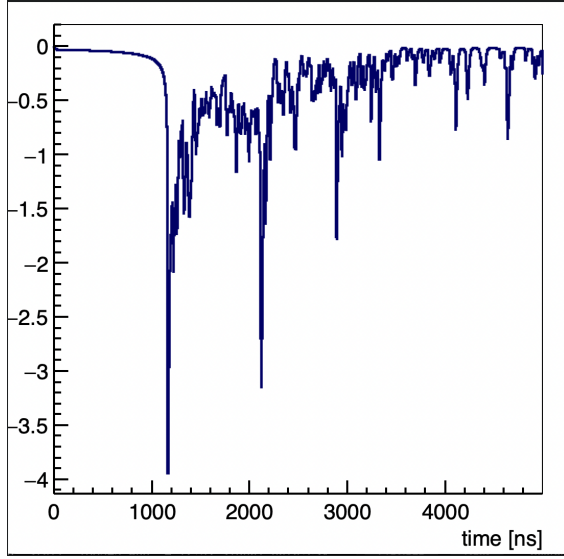


FIG. 4. Signal plot produced by Garfield++.

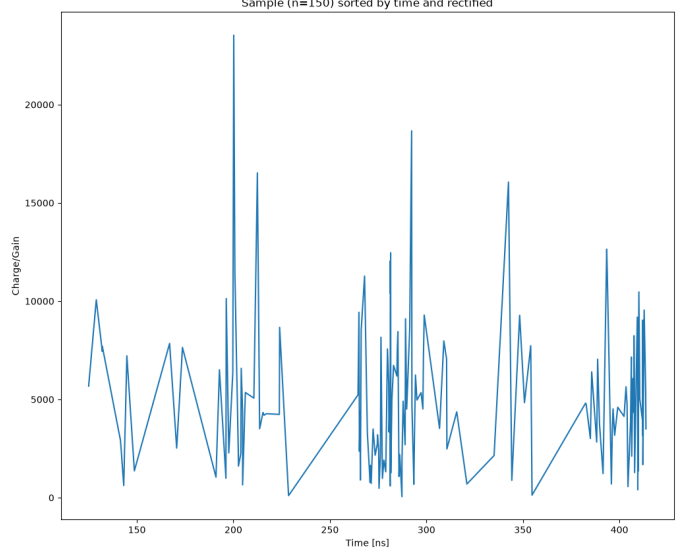


FIG. 5. Signal plot produced by Python using reconstructed data.

In *DriftChamberSim*, the shape of a signal pulse is modelled by

$$f(t) = \frac{t}{\tau} e^{-t/\tau} \quad (2)$$

with a peak response time of $\tau = 8$ ns. Garfield++ natively includes signal plotting tools (see FIG 4.), which were recreated in Python to handle the reconstructed data. This was done by first estimating the number of clusters over the length of the drift from a Poisson distribution using $\lambda = \mu \ell$. The mean number of clusters as estimated to be $\mu = 109.69$ clusters/cm by averaging the number of clusters between $\frac{dN}{dx} = 110$ and $\frac{dN}{dx} = 315$ across $n = 2000$ tracks in Garfield++. Tracks with $\frac{dN}{dx} > 315$ were generally observed to be associated with delta rays, while tracks with $\frac{dN}{dx} < 110$ were generally observed to be a result of insignificantly small track lengths, and these served as the cut boundaries for the sample (see Fig 2.).

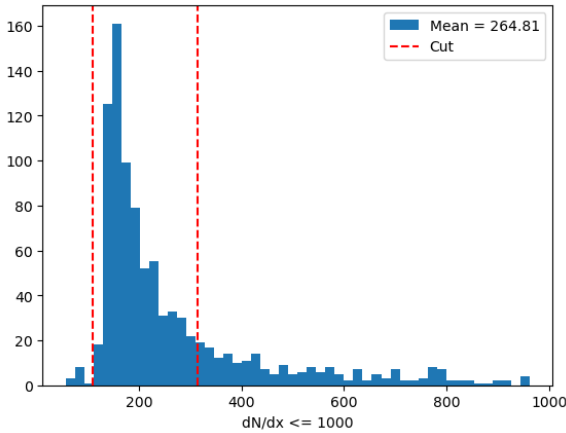
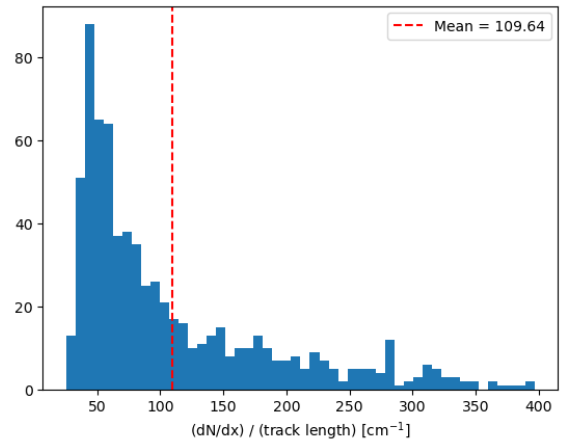


FIG. 6. Distribution of the number of clusters for tracks with less than 1000 clusters.

FIG. 7. Clusters per cm of track for tracks with $\frac{dN}{dx} \in [110, 315]$.

The position of each particle along the track within the cell was uniformly distributed. The drift time of each electron was smeared using the response function and the combined quadrature standard deviation of the transverse and longitudinal diffusion coefficients, derived from the lookup tables generated by *DriftTimes*. The gain was randomly sampled from a Pólya probability density function, effectively a gamma distribution, with a mean gain $G = 20000$ and a shape parameter of $m = 10$ to produce a signal plot of the simulation data (see Fig 5.). Comparing the signal plots produced by Garfield++ (see Fig 4.) and Geant4 (see Fig 5.), differences in the data scale and resolution are apparent, but similar clustering behaviour is observed.

SUGGESTED IMPROVEMENTS

DriftChamberSim is still in active development and is working to improve the efficiency of its programming, as well as the accuracy of its data collection. Below are details about future changes that would likely be effective at reaching this goal:

- Adding the missing endplate within the Z_1 to Z_2 region would allow the model to more closely resemble the true structure of the CDC. Additionally, offering an option to exclude these endplates would allow researchers to determine if they are even necessary, as per another upgrade proposal.
- Currently, *DriftChamberSim* models the shape of the signal pulse using an arbitrary assumed peak response time of $\tau = 8$ ns. Readout electronics like in the CDC have inherent bandwidth limits and integration times. Finding the smallest viable τ helps model the fastest possible response the physical electronics could achieve without completely missing the signal. A more accurate value could be determined by testing various peak response times in Garfield++ and determining the smallest response time with which a signal response can still be observed.
- To improve the accuracy of the reconstruction to the simulation data, testing should be conducted to compare the effectiveness of GenFit's Deterministic Annealing Filter (DAF) fitter to the Kalman fitter currently used. This DAF fitter is a modified version of the Kalman fitter that weights the inputs to handle outliers in the data and resolves detector ambiguities.

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APPENDIX I

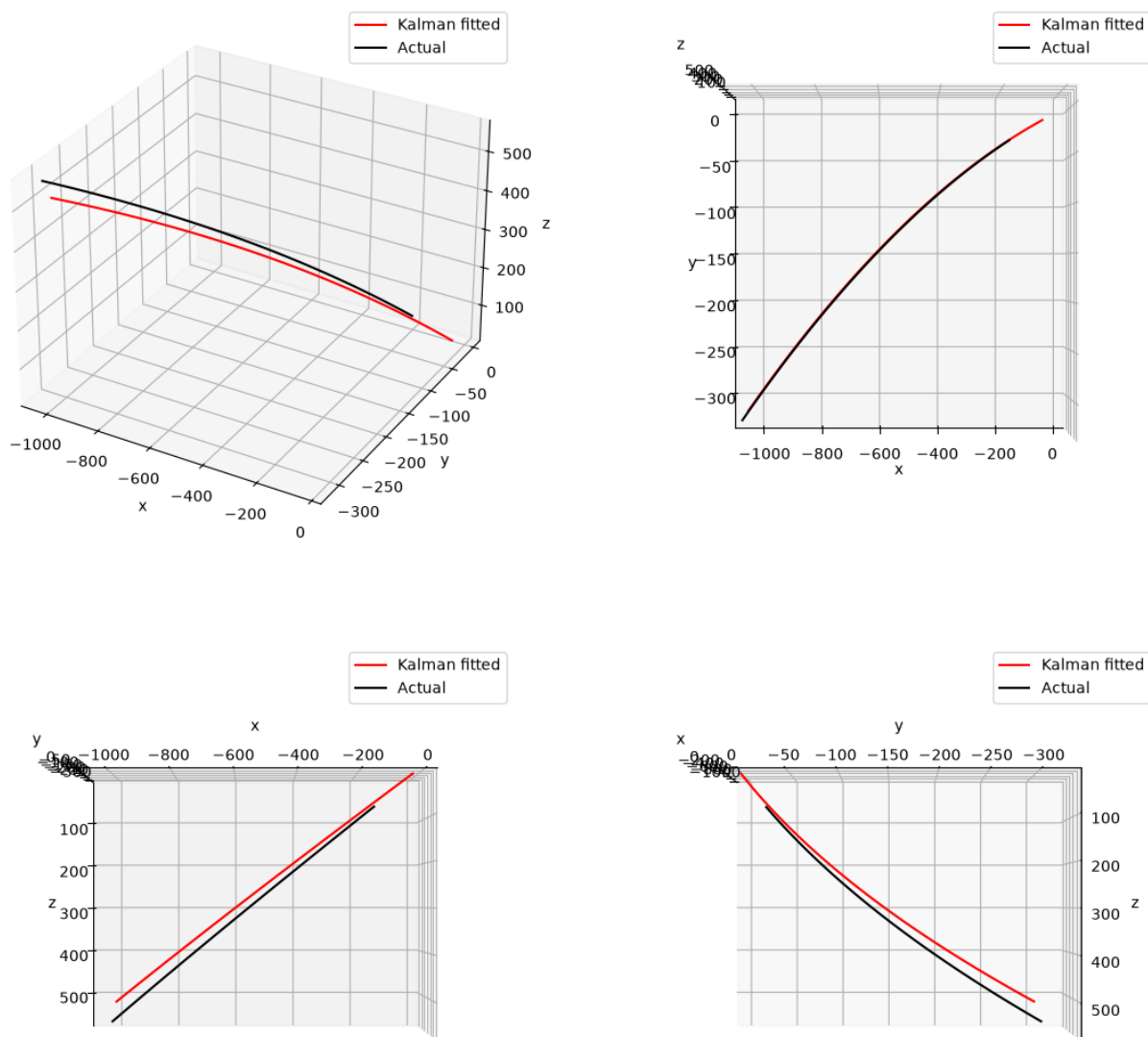


FIG. 8. Comparison between the simulated and reconstructed particle trajectories, with distances in mm.